

Notes: Becker & Ivashina (JF, 2015)

Reaching for Yield in the Bond Market

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Data construction and measurement	4
3.1	Insurance companies and institutional bond holdings	4
3.2	Bond characteristics and ratings	4
3.3	Yield spreads and CDS spreads	4
3.4	Secondary-market trading and returns	5
3.5	Insurance-company characteristics	5
4	Empirical specifications and model logic	5
4.1	Benchmark decomposition of yield spreads	5
4.2	Primary-market holdings regression	6
4.3	Secondary-market trading regression	6
4.4	Time-series variation in reaching	6
4.5	Performance, downgrade, and insurer outcome tests	7
4.6	Governance and capital regressions	7
5	Identification strategy and interpretation	7
5.1	Core identification idea: risk within regulatory categories	7
5.2	Why comparing investor types matters	7
5.3	Why primary-market tests are useful	7
5.4	Why secondary-market tests strengthen the evidence	8
5.5	Alternative explanations and how the paper addresses them	8
5.6	Remaining threats to interpretation	8
6	Tables and figures	8
6.1	Figure 1: Insurance companies' holdings of newly issued bonds	8
6.2	Table I: Distribution of Assets and Table II: NAIC Risk-Based Capital Requirements	9
6.3	Figure 2: Newly issued U.S. corporate bonds and Table III: Bond and issuer characteristics across CDS quartiles	10
6.4	Figure 3: Reaching for yield and Tables IV–V: Primary market yield and CDS spread regressions	11
6.5	Figure 4: Relation between CDS spread and yield to maturity and Figure 5: Insurance companies' capital structure	13
6.6	Table VI: Reaching for yield in secondary-market trading and Table VII: Time-series variation	13
6.7	Table VIII: Bond performance return analysis and Table IX: Bond performance downgrades .	14

6.8	Table X: Governance and reaching for yield and Table XI: Reaching for yield and insurers' performance	15
6.9	Figure 6: Insurance companies' stock price performance during the financial crisis and Figure 7: Reaching for yield and bond issuance	16
7	Interpretive synthesis	17
8	Short summary	17
9	Conclusion	18
9.1	Core conclusion	18
9.2	Economic intuition	18
9.3	What the paper establishes	18
9.4	What the paper does not fully establish	18
9.5	Final takeaway	18

1 Big picture

- **Question.** Do institutional investors take extra risk within regulatory and contractual risk categories in order to earn higher yields?
- **Main setting.** The paper studies U.S. insurance companies, the largest institutional holders of corporate and foreign bonds, and asks whether they buy riskier bonds within the same NAIC/rating category.
- **Core idea.** When regulation and delegated investment contracts use imperfect risk metrics such as credit ratings, managers can improve apparent yield without violating the metric by buying bonds that are riskier on dimensions not fully captured by ratings.
- **Definition of reaching for yield.** Reaching means overweighting bonds that offer higher yield spreads or CDS spreads conditional on the same rating/regulatory category.
- **Main result.** Insurance companies prefer higher-yielding and higher-CDS-spread bonds relative to mutual funds and pension funds, especially in the pre-crisis boom period.
- **Mechanism.** The behavior is stronger when regulatory capital is more binding and when governance is weaker, suggesting both regulatory incentives and agency frictions.
- **Time-series pattern.** Reaching is procyclical: it is pronounced in 2004–2007, disappears during the crisis, and reappears after markets stabilize.
- **Performance result.** Bonds heavily held by insurers do not earn positive alpha; if anything, they have negative abnormal returns and more downgrades. Insurers that reached more pre-crisis performed worse during the 2008–2009 crisis.
- **Real/issuance implication.** High-CDS BBB issuers issue more when insurance reaching is stronger, suggesting that reaching can affect the allocation of corporate credit.
- **Takeaway.** The paper provides micro evidence that imperfect risk measures can create incentives for delegated institutions to buy “safe-looking” but risky assets, amplifying credit booms and exposing regulated intermediaries to hidden risk.

2 Incremental contribution of the paper

- **From theory and anecdotes to micro evidence.** Before this paper, reaching for yield was often discussed as a narrative explanation for credit booms. Becker and Ivashina provide security-level evidence from institutional portfolios.
- **Separates rating risk from non-rating risk.** The key empirical move is to compare bonds *within the same NAIC/rating category*. This isolates risk that is not priced by the regulatory capital metric but is visible in yields and CDS spreads.
- **Uses investor-type comparisons.** The paper compares insurance-company holdings to mutual-fund and pension-fund holdings of the same issuance universe. This helps show that the tilt is specific to insurers rather than a generic feature of all investors.
- **Connects asset choice to incentives.** The paper does not simply document high-yield holdings. It links reaching to low regulatory capital surplus and weak blockholder governance, providing evidence on why the behavior arises.
- **Tests the “alpha” alternative.** If insurers bought high-yield bonds because they were skilled investors, high-insurance portfolios should outperform. Instead, return and downgrade tests show no positive alpha and more downside risk.
- **Links portfolio distortion to broader credit supply.** The paper goes beyond portfolio choice by showing suggestive links between reaching and corporate bond issuance by high-CDS firms within the BBB category.

- **Broad contribution.** The article bridges regulation, delegated portfolio management, fixed-income asset pricing, and credit-supply cyclicality. Its central lesson is that risk-based regulation can itself create risk-taking incentives when the risk measure is incomplete.

3 Data construction and measurement

3.1 Insurance companies and institutional bond holdings

- The main institutional holdings data come from eMAXX, which reports corporate bond holdings by institutional investor.
- The dataset identifies investor type, including insurance companies, mutual funds, and pension funds.
- eMAXX contains manager-level and ultimate-investor-level information. This matters because bond positions can be managed by an external manager but ultimately owned by an insurer.
- The paper focuses on insurance companies because insurers are large corporate bond investors and are subject to rating-based capital requirements.
- The holdings sample covers more than four million individual corporate bond holdings, with reported face value around \$1–1.4 trillion depending on the quarter.
- The new-issue sample contains 3,709 bonds issued by 320 firms from 2004:Q3 to 2010:Q4.

Interpretation: The key advantage of eMAXX is that it reveals who buys each bond at issuance and how holdings change over time. This allows the authors to detect reaching directly from investor portfolios rather than inferring it from prices alone.

3.2 Bond characteristics and ratings

- Bond characteristics are from Mergent FISD.
- The paper excludes convertibles, preferred stock, other preferred securities, government bonds, and government-sponsored enterprise bonds.
- Floating-rate bonds are excluded because promised yields are not directly comparable.
- Credit ratings are taken from S&P, Moody’s, and Fitch and converted into NAIC categories.
- If a bond has two ratings, the lower rating is used; if it has three ratings, the median rating is used.
- NAIC 1 includes AAA, AA, and A; NAIC 2 includes BBB; NAIC 3–6 are progressively lower high-yield categories.

Interpretation: The rating-to-NAIC mapping is central. Capital requirements change discretely across NAIC categories but not within each category. Therefore, within-category variation in yield or CDS spread is the space where reaching can occur without immediately raising regulatory capital charges.

3.3 Yield spreads and CDS spreads

- Yield spread is the promised yield at issuance minus the yield on a duration-matched Treasury bond.
- When Mergent FISD does not report a spread, the authors infer it from yield curves and bond maturity.
- CDS spreads come from Credit Market Analysis through Datastream and use five-year senior CDS contracts.

- CDS spreads are an alternative market-based measure of credit risk that may capture default risk and priced risk not reflected in ratings.
- The CDS sample is restricted to issuers included in CDX Markit index series, which helps ensure liquidity but focuses on larger and more traded issuers.

Interpretation: Yield spreads and CDS spreads are the two main proxies for risk not fully captured by ratings. If insurers buy higher-yield or higher-CDS bonds within the same rating category, they are taking more risk while staying inside the same regulatory bucket.

3.4 Secondary-market trading and returns

- Secondary-market trades are from TRACE.
- Trades are cleaned following standard procedures, excluding canceled/corrected trades and duplicate transactions close in time.
- Monthly returns are constructed from median transaction prices on the last active trading day of each month, including coupons.
- The return tests use large trades to focus on institutional activity and avoid retail-trade noise.
- Return factors include stock market excess returns, the default premium, the term spread, and the Pástor–Stambaugh liquidity factor.

Interpretation: These data allow the paper to test whether high-insurance holdings reflect skill. If insurers are buying underpriced bonds, such bonds should earn positive abnormal returns. The paper finds the opposite.

3.5 Insurance-company characteristics

- Regulatory capital surplus is collected from S&P Capital IQ and measured as reported capital minus required capital, normalized by assets.
- When a subsidiary appears in the data, the ultimate parent’s capital is used.
- Public listing status, leverage, size, stock beta, stock volatility, and crisis returns are collected for publicly listed insurers.
- Governance is proxied by blockholder ownership, updated from Dlugosz et al. using 13F filings.
- An insurer’s propensity to reach is measured by the equal-weighted average yield spread of newly acquired investment-grade bonds.

Interpretation: The insurer-level variables help distinguish mechanisms. If reaching is driven by regulation, capital-constrained insurers should reach more. If it is driven by agency problems, stronger blockholders should be associated with less reaching.

4 Empirical specifications and model logic

4.1 Benchmark decomposition of yield spreads

The paper starts from the idea that a bond’s yield spread can be decomposed as

$$Y_i - R_f \approx PD_i \times LGD_i + RP_i + \alpha_i, \quad (1)$$

where PD is default probability, LGD is loss given default, RP is the risk premium, and α is abnormal return. A rating-based benchmark captures only a coarse estimate:

$$Y_i - R_f = PD_i^{CR} \times LGD_i^{CR} + [(PD_i LGD_i - PD_i^{CR} LGD_i^{CR}) + RP_i] + \alpha_i. \quad (2)$$

Interpretation: The bracketed term is the source of reaching. If the manager is judged against the rating-based benchmark, then taking more priced risk not captured by the rating can raise yield without obviously violating the constraint.

4.2 Primary-market holdings regression

The primary-market specification is conceptually

$$InsuranceShare_{b,t} = \alpha + \beta Spread_{b,t} + \Gamma' X_{b,t} + FE + \varepsilon_{b,t}, \quad (3)$$

where $InsuranceShare$ is the share of institutional holdings held by insurance companies relative to insurers, mutual funds, and pension funds.

- $Spread$ is either yield spread or CDS spread.
- Controls include duration, issue size, trading volume, and number of investors/trades.
- Some specifications include interacted rating-by-year fixed effects.
- The test is run separately for pre-crisis and crisis/post-crisis periods.

Interpretation: A positive β within a rating/NAIC category means insurers buy more of the high-yield or high-CDS bonds relative to other investors. That is the core reaching-for-yield test.

4.3 Secondary-market trading regression

The secondary-market specification is

$$\Delta \log Holdings_{b,t}^{Ins} = \alpha_{firm,t} + \beta Spread_{b,t} + \Gamma' X_{b,t} + u_{b,t}. \quad (4)$$

- The dependent variable is the quarterly log change in insurance holdings of a bond.
- Firm-quarter fixed effects compare bonds from the same issuer at the same time.
- Yield spread is measured relative to a Treasury maturity match.

Interpretation: This test is stronger than a cross-sectional purchase test because it follows changes in holdings over time and absorbs issuer-quarter conditions. A positive coefficient means insurers add more to high-yield bonds even within the same issuer-quarter environment.

4.4 Time-series variation in reaching

The paper estimates quarter-specific coefficients:

$$\Delta \log Holdings_{b,t}^{Ins} = \alpha_{firm,t} + \sum_q \beta_q (Spread_{b,t} \times 1\{t = q\}) + \Gamma' X_{b,t} + u_{b,t}. \quad (5)$$

Interpretation: The β_q coefficients show when reaching is strongest. Positive coefficients in expansions and weak or negative coefficients in crisis quarters support the idea that reaching is cyclical and linked to credit booms.

4.5 Performance, downgrade, and insurer outcome tests

The paper tests whether reaching reflects skill or risk by estimating bond return and downgrade regressions and insurer crisis-return regressions:

$$R_{p,t}^{excess} = \alpha_p + \beta' Factors_t + \varepsilon_{p,t}, \quad (6)$$

$$Downgrade_{b,t+h} = \alpha + \delta InsuranceHoldings_{b,t} + \Gamma' X_{b,t} + e_{b,t+h}, \quad (7)$$

$$CrisisReturn_i = \alpha + \theta AvgYieldSpread_i + \Gamma' Z_i + \eta_i. \quad (8)$$

Interpretation: If insurers are skillfully identifying undervalued bonds, high-insurance bonds should have positive alpha and not be downgraded more. Instead, the paper finds negative alphas, higher downgrade probability, and worse insurer crisis performance.

4.6 Governance and capital regressions

The insurer-level incentive test is

$$AvgYieldSpread_{i,t} = \alpha + \beta_1 CapitalSurplus_{i,t} + \beta_2 BlockholderOwnership_{i,t} + \Gamma' Z_{i,t} + \varepsilon_{i,t}. \quad (9)$$

Interpretation: A negative coefficient on capital surplus means insurers with less spare regulatory capital buy higher-spread bonds. A negative coefficient on blockholder ownership means stronger governance reduces reaching.

5 Identification strategy and interpretation

5.1 Core identification idea: risk within regulatory categories

The central identification strategy is to compare bonds that look similar to regulation but differ in market-implied risk. NAIC capital rules use coarse rating categories. Within a NAIC category, two bonds may have the same regulatory capital charge but very different yields, CDS spreads, liquidity, or risk premia. If insurers overweight the high-yield bond within the same category, they are exploiting the gap between regulatory risk measurement and market risk pricing.

5.2 Why comparing investor types matters

The main outcome is not simply whether insurers buy high-yield bonds, but whether insurers buy more than other institutional investors. Mutual funds and pension funds serve as benchmark investors. This comparison removes the possibility that high-yield bonds are simply attractive to everyone. The identified pattern is specific to insurance companies, whose capital requirements and institutional incentives are tied to ratings.

5.3 Why primary-market tests are useful

Primary-market allocations capture investors' choices at issuance, before many secondary-market trading confounds can arise. The test asks whether insurers take a larger share of an issue when the yield is higher conditional on the same NAIC/rating bucket. Controls for issue size, maturity, liquidity, and rating-by-year fixed effects reduce the concern that high-yield bonds differ mechanically on observables.

5.4 Why secondary-market tests strengthen the evidence

Secondary-market trading tests use changes in insurance holdings and can include firm-quarter fixed effects. This means the comparison is among bonds of the same issuer in the same quarter. The test is therefore less vulnerable to unobserved issuer quality. It asks whether insurers increase positions in the higher-yield securities of the same issuer-time environment.

5.5 Alternative explanations and how the paper addresses them

- **Liquidity preference.** The paper controls for trading volume and number of trades/investors. Reaching persists in key pre-crisis specifications.
- **Investment skill.** Return analysis shows high-insurance bonds do not earn positive abnormal returns; their alphas are negative.
- **Unobserved default risk.** Downgrade tests show high-insurance holdings predict more downgrades, consistent with hidden risk rather than safe excess yield.
- **General high-yield demand.** Comparisons to mutual funds and pension funds show the pattern is concentrated among insurers.
- **Regulation and agency mechanisms.** Capital surplus and blockholder ownership tests connect reaching to institutional incentives.

5.6 Remaining threats to interpretation

The paper does not randomly assign investors to capital constraints or bonds to yields. CDS coverage focuses on large issuers. eMAXX does not cover all investor types, and portfolio returns are difficult to measure perfectly in illiquid bond markets. Therefore, the strongest interpretation is that the paper provides a tightly documented pattern and mechanism-consistent evidence rather than a fully randomized causal experiment.

6 Tables and figures

6.1 Figure 1: Insurance companies' holdings of newly issued bonds

Read:

- Panel A shows insurers are much more likely to hold investment-grade bonds, especially NAIC 1 and 2, than lower-rated categories.
- Panel B focuses on NAIC 1 and shows that, within the same safe regulatory category, insurers hold more of bonds with high yield spreads and high CDS spreads.
- This is the visual version of the main reaching result.

Interpretation: The figure separates two forces. First, regulation pushes insurers toward high ratings. Second, conditional on high ratings, insurers move toward higher yield or higher CDS risk. Reaching occurs inside the safe-looking regulatory bucket.

Economic message: Credit ratings reduce but do not eliminate risk-taking incentives. The relevant distortion is not buying junk bonds; it is buying the riskiest bonds that still qualify as safe.

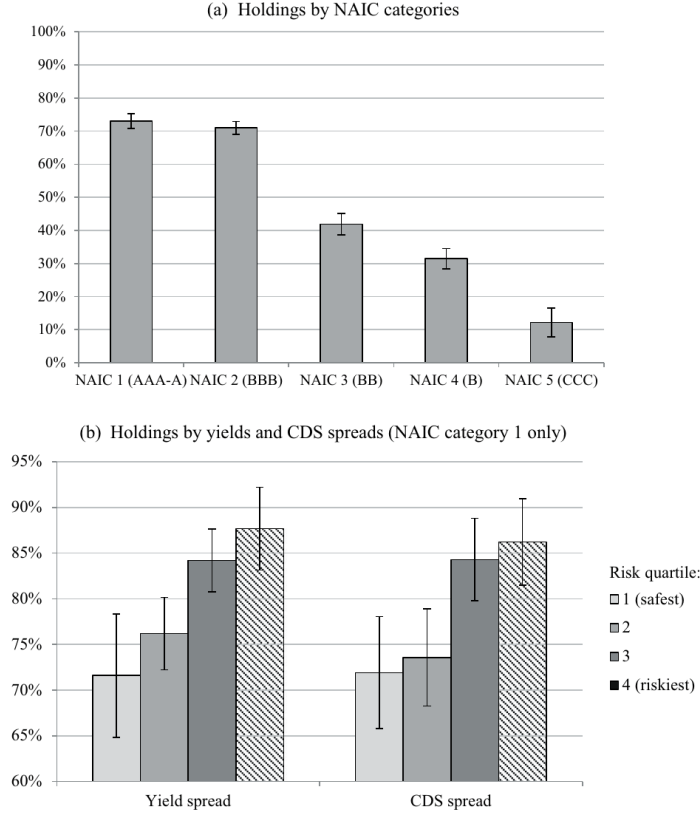


Figure 1. Insurance companies' holdings of newly issued bonds. This figure plots the fraction of newly issued bonds acquired by insurance companies between 2004:Q2 to 2007:Q2, sorted by NAIC risk categories (panel A), as well as promised yields and CDS spreads (panel B). The fraction of holdings is computed with respect to the total dollar holdings of insurance companies, pension funds, and mutual funds. We report equal-weighted averages across bonds. Bars correspond to 95% confidence intervals (based on the standard deviation across individual bonds). In panel A, bonds are sorted by NAIC categories (except NAIC category 6, which has few issues). In panel B, investment grade bonds (NAIC categories 1 and 2) are sorted into quartiles of duration.

6.2 Table I: Distribution of Assets and Table II: NAIC Risk-Based Capital Requirements

Read:

- Table I shows that life insurers hold about \$2 trillion in corporate and foreign bonds in 2010:Q4, equal to 39% of their total financial assets.
- Property/casualty insurers also have large bond exposures, though municipal securities are more important for them.
- Table II shows the regulatory capital charge rises from 0.3% for NAIC 1 to 0.96% for NAIC 2, 3.39% for NAIC 3, and much higher for lower categories.
- The capital charge jumps discretely across categories, while yield and CDS spreads vary continuously within categories.

Interpretation: The tables establish both economic importance and institutional incentives. Insurers are major bond buyers, and their regulation uses ratings as coarse risk buckets. This is exactly the environment in which reaching within a rating category can arise.

Economic message: The institutional design makes a high-yield BBB bond appealing: it offers a higher spread without moving outside the same NAIC 2 capital bucket.

Table I
Distribution of Assets

This figure is based on U.S. Flow of Funds Accounts. The numbers are in billions of U.S. dollars and correspond to amounts outstanding as of the end of 2010:Q4. Mutual funds include money market funds.

	Life Insurance Companies	Property/Casualty Insurance Companies	Mutual Funds	Private Pension Funds
Total financial assets	5,176.8	1,403.4	7,963.4	6,079.6
Treasury securities	161.6 (3%)	91.4 (7%)	296.0 (4%)	486.7 (8%)
Corporate and foreign bonds	2,022.7 (39%)	299.2 (21%)	1,255.0 (16%)	482.5 (8%)
Agency- and GSE-backed securities	357.0 (7%)	109.4 (8%)	786.7 (10%)	170.9 (3%)
Mortgages	318.0 (6%)	4.1 (0%)	—	15.1 (0%)
Municipal securities and loans	77.5 (1%)	369.8 (26%)	525.9 (7%)	—
Corporate equities	1,423.2 (27%)	228.0 (16%)	4,801.4 (60%)	1,983.3 (33%)
Mutual fund shares	183.8 (4%)	32.1 (2%)	—	2,228.2 (37%)
Total	88%	81%	66%	88%

(a) Table I: Distribution of assets

Table II
NAIC Risk-Based Capital Requirements

This table summarizes National Association of Insurance Companies (NAIC) post-tax capital requirement factors (NAIC Risk-Based Capital Newsletter, 10/12/2001). Default rates are from Fitch Ratings Global Corporate Finance 2010 Transition and Default Study.

NAIC Category	Credit Ratings	Capital Charge	Five-Year Cumulative Default Rates (1990–2010)
Federal government			
NAIC 1 (highest)	AAA, AA, A	Investment grade	0.3%
NAIC 2	BBB	Investment grade	0.96%
NAIC 3	BB	High yield/speculative grade	6.76%
NAIC 4	B	High yield/speculative grade	7.38%
NAIC 5	CCC	High yield/speculative grade	16.96%
NAIC 6 (lowest)	CC or below	High yield/speculative grade	19.50%

(b) Table II: NAIC capital requirements

6.3 Figure 2: Newly issued U.S. corporate bonds and Table III: Bond and issuer characteristics across CDS quartiles

Read:

- Figure 2 shows that most new corporate bond issuance in the sample is investment grade, especially NAIC 1 and NAIC 2.
- Table III sorts investment-grade bonds by CDS spread quartile.
- The riskiest CDS quartile has much higher yield spreads and CDS spreads but only a slightly lower average credit rating.
- Riskier CDS issuers have higher leverage and lower return on assets.

Interpretation: The table validates the identifying variation. Within investment grade, CDS spreads identify economically meaningful risk differences that ratings do not fully capture. High-CDS issuers are worse on fundamentals even though ratings remain close.

Economic message: Ratings are too coarse to summarize all priced credit risk. This gap gives insurers room to reach for yield while staying inside rating constraints.

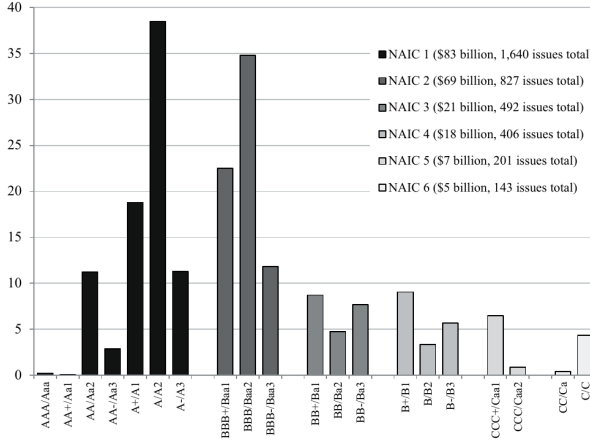


Figure 2. Newly issued U.S. corporate bonds, 2004:Q3 to 2010:Q4.

(a) Figure 2: New issue distribution

Table III
Bond and Issuer Characteristics across CDS Quartiles
(Investment-Grade Bonds)

This table presents the median of selected issuer and bond characteristics over the sample period for investment-grade issuers. The characteristics are at bond issuance. Each entry reports the median and, in parentheses, standard deviation of a characteristic. For credit ratings, the standard deviation is in notches (the difference between A and A- is one notch).

Quartile	Yield Spread	Maturity	CDS Spread	Credit Rating	Assets (Billion USD)	Book Leverage	Return on Assets
1 (safest)	65.4 (21.1)	9.8 (13.2)	16.9 (14.1)	A (2.3)	25.7 (120)	0.24 (0.13)	0.067 (0.047)
2	112.3 (22.0)	9.0 (11.5)	43.5 (18.8)	A (1.8)	21.5 (125)	0.28 (0.16)	0.053 (0.052)
3	170.9 (24.7)	10.0 (12.1)	57.8 (32.6)	A- (1.8)	15.4 (167)	0.30 (0.16)	0.044 (0.061)
4 (riskiest)	344.2 (124.3)	9.2 (9.6)	156.8 (287.5)	A- (1.8)	13.7 (170)	0.31 (0.20)	0.022 (0.085)

(b) Table III: CDS quartiles

6.4 Figure 3: Reaching for yield and Tables IV–V: Primary market yield and CDS spread regressions

Read:

- Figure 3 shows insurers hold higher-yielding NAIC 1 bonds than pension funds and mutual funds in the pre-crisis period.
- During the crisis/post-crisis period, the pattern reverses or weakens.
- Table IV shows positive pre-crisis coefficients on yield spread: insurers hold a larger fraction of high-yield bonds at issuance.
- Table V shows similar evidence using CDS spreads, especially before adding the most demanding fixed effects.
- In the crisis period, the yield/CDS reaching pattern becomes negative or disappears.

Interpretation: These are the central tests. Conditional on ratings and controls, insurers disproportionately buy high-yield investment-grade bonds before the crisis. The weakening during the crisis supports a cyclical interpretation: reaching is a boom phenomenon rather than a mechanical preference for high-yield bonds in all periods.

Economic message: Reaching for yield is strongest when credit conditions are benign and risk appears cheap. This is exactly when a buildup of hidden risk is most dangerous.

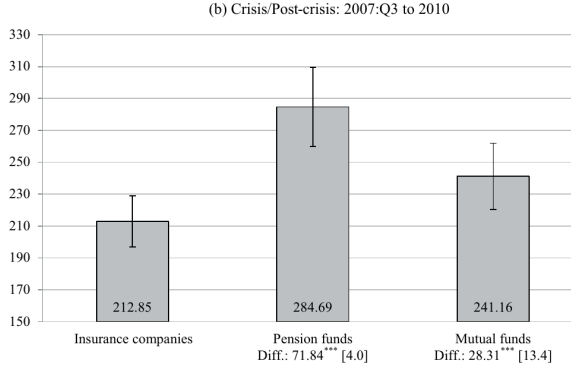
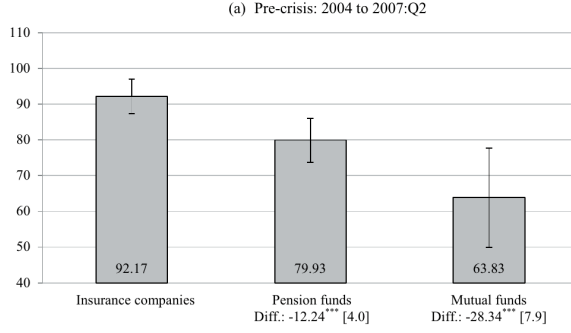


Table IV
Reaching for Yield: Primary Market, Yield Spreads

This table looks at the investment decision immediately following bond issuance. With the exception of specification 4, the sample comprises investment-grade issues (NAIC categories 1 and 2) specification 4 corresponds to noninvestment/speculative grade. *Yield spread* is the yield difference between a bond and a Treasury. *Duration* is computed using bond information from Mergent FISD. *Trading volume* is the log of the value of transactions in TRACE in the first quarter of trading divided by total par value outstanding. *Number of trades* is the log of the number of investors reporting a change in holdings in the first quarter of trading. Credit rating fixed effects are AAA, AA+, AA-, etc. Standard errors are clustered by issuer and reported in brackets. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

Dependent Variable	Insurance Companies' Holdings as a Fraction of Total Holdings of Insurance Companies, Mutual Funds, and Pension Funds						
	Benchmark (2004:Q3 -2007:Q2)				Crisis (2007:Q3 -2010:Q4)		
	Investment Grade		Speculative		Investment Grade		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Yield spread	0.054* [0.029]	0.074*** [0.025]	0.048** [0.023]	0.030 [0.019]	-0.036* [0.016]	-0.051*** [0.014]	-0.043*** [0.017]
NAIC category 2 (BBB+, BBB, BBB-)	-0.006 [0.051]	-0.010 [0.022]	- [0.004]	- [0.009]	-0.055 [0.070]	0.055 [0.033]	- [0.008]
Duration (years)	- [0.003]	0.007** [0.003]	0.008** [0.004]	0.017* [0.009]	- [0.008]	0.028 [0.008]	0.003 [0.005]
Offering amount	- [0.005]	-0.006 [0.005]	-0.002 [0.006]	-0.060*** [0.016]	- [0.028]	-0.000 [0.028]	-0.026 [0.030]
Trading volume	- [0.011]	-0.022** [0.011]	-0.015 [0.012]	-0.002 [0.021]	- [0.018]	-0.031* [0.018]	-0.041** [0.018]
Number of trades	- [0.018]	0.007 [0.018]	-0.005 [0.023]	0.023 [0.027]	- [0.026]	-0.022 [0.026]	0.018 [0.029]
Fixed effects: rating × year (interacted)	No	No	Yes	Yes	No	No	Yes
Observations	600	468	468	190	373	304	304
Clusters (issuers)	141	137	137	84	104	100	100

Table V
Reaching for Yield: Primary Market, CDS Spreads

This table looks at the investment decision immediately following bond issuance. With the exception of specification 4, the sample comprises investment-grade issues (NAIC categories 1 and 2) specification 4 corresponds to noninvestment/speculative grade. *Duration* is computed using bond information from Mergent FISD. *Trading volume* is the log of the value of transactions in TRACE in the first quarter of trading divided by total par value outstanding. *Number of trades* is the log of the number of investors reporting a change in holdings in the first quarter of trading. Credit rating fixed effects are AAA, AA+, AA-, etc. Standard errors are clustered by issuer and reported in brackets. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

Dependent Variable	Insurance Companies' Holdings as a Fraction of Total Holdings of Insurance Companies, Mutual Funds, and Pension Funds						
	Benchmark (2004:Q3 -2007:Q2)				Crisis (2007:Q3 -2010:Q4)		
	Investment Grade		Speculative		Investment Grade		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
CDS spread	0.016* [0.008]	0.018** [0.007]	0.001 [0.001]	-0.001 [0.001]	-0.001*** [0.000]	-0.001*** [0.000]	-0.001** [0.000]
NAIC category 2 (BBB+, BBB, BBB-)	-0.070 [0.048]	-0.046 [0.039]	- [0.011]	- [0.009]	-0.058 [0.060]	0.040 [0.035]	- [0.006]
Duration (years)	- [0.005]	0.011** [0.005]	0.011* [0.006]	0.009 [0.008]	- [0.004]	0.008** [0.004]	0.006 [0.004]
Offering amount	- [0.017]	-0.013 [0.012]	0.003 [0.012]	-0.060*** [0.020]	- [0.026]	-0.013 [0.026]	-0.046 [0.030]
Trading volume	- [0.014]	-0.029** [0.014]	-0.024* [0.013]	-0.003 [0.024]	- [0.015]	-0.031** [0.015]	-0.035** [0.016]
Number of trades	- [0.022]	0.022 [0.022]	0.012 [0.024]	0.033 [0.027]	- [0.025]	0.003 [0.025]	0.024 [0.029]
Fixed effects: rating × year (interacted)	No	No	Yes	Yes	No	No	Yes
Observations	453	360	360	174	348	280	280
Clusters (issuers)	118	115	115	32	99	35	35
R ²	0.02	0.06	0.26	0.42	0.11	0.20	0.32

6.5 Figure 4: Relation between CDS spread and yield to maturity and Figure 5: Insurance companies' capital structure

Read:

- Figure 4 shows a positive but imperfect relation between CDS spreads and bond yield spreads.
- The imperfect correlation matters because the two measures capture related but not identical dimensions of risk and market pricing.
- Figure 5 shows that insurance liabilities dominate insurer capital structures.
- Equity is a relatively small share of total financing for insurers, making risk-taking within portfolios economically important.

Interpretation: Figure 4 justifies using both yield and CDS measures. Figure 5 helps explain why portfolio reaching can have systemic implications: insurers have large fixed liabilities and relatively thin equity buffers.

Economic message: When a leveraged insurer reaches for yield, the hidden downside can show up as losses to a broad base of policyholders and investors during downturns.

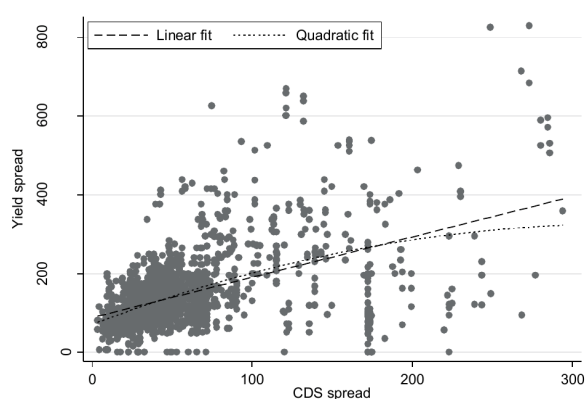


Figure 4. Relation between CDS spread and yield to maturity at bond issuance. This plotted sample corresponds to NAIC risk categories 1 and 2 and excludes CDS spreads in excess of 300 bps. The main dependent variable in our analysis (insurance bond holdings) is quarterly. The

(a) Figure 4: CDS and yield spreads

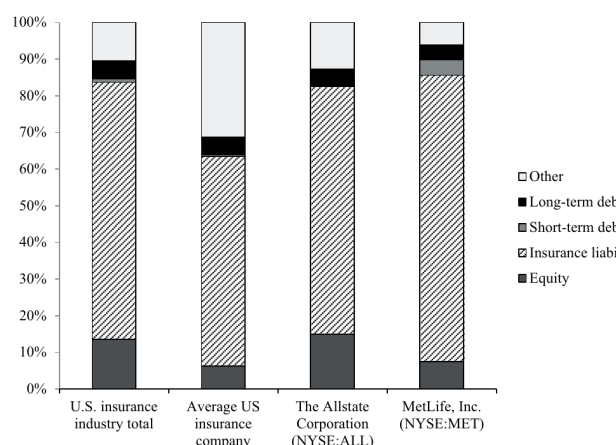


Figure 5. Insurance companies' capital structure. The figure presents capital structure data for 17 U.S. insurance companies with more than \$100 million of assets for fiscal year 2010

(b) Figure 5: Insurer capital structure

6.6 Table VI: Reaching for yield in secondary-market trading and Table VII: Time-series variation

Read:

- Table VI shows insurers increase secondary-market holdings of high-yield investment-grade bonds in the benchmark period.
- The dependent variable is the log change in insurer holdings, so the result is about active portfolio adjustment rather than only initial allocation.
- Table VII lets the yield-spread coefficient vary by quarter.
- Positive coefficients appear in many pre-crisis quarters, disappear around the crisis, and reappear in some post-crisis quarters.

Interpretation: The secondary-market test addresses concerns that issuance allocations mechanically differ across investors. The time-series test shows reaching is state dependent. It rises in good times and weakens in stress periods.

Economic message: Reaching is not a fixed taste. It is part of a credit-cycle mechanism: risk-taking grows during booms and collapses during stress.

Table VI
Reaching for Yield: Trading in the Secondary Market

This table reports results from panel regressions of secondary market activity of insurance companies in investment-grade corporate bonds. The dependent variable is the log quarterly change in the value of all insurance companies' holdings of a bond (changes in holdings due to a bond's maturity are excluded). Observations for which the dependent variable is above one are excluded. Similarly, bonds for which total holdings are never above \$10 million are excluded. The sample is constrained to investment-grade issues (NAIC categories 1 and 2). *Yield spread* is the yield maturity at which the bond has last transacted minus the end-of-quarter yield on the Treasury matched on duration. *Trading volume* is the log of the value of transactions in TRACE for a given bond, divided by total par value outstanding, lagged one quarter. *Number of trades* is the log of the number of investors reporting a changed position in the previous quarter in eMAXX. Standard errors are clustered by issuer and reported in brackets. *** and ** indicate statistical significance at the 1% and 5% level, respectively.

Dependent variable:	Log Change in the Value of Insurance Companies' Total Holdings Since the Previous Quarter		
	Benchmark (2004:Q3–2007:Q2)		Crisis (2007:Q3–2010:Q4)
	(1)	(2)	(3)
Yield spread	0.025** [0.010]	0.024** [0.010]	0.000 [0.001]
NAIC category 2 (BBB+, BBB, BBB–)	–0.069 [0.049]	–0.017*** [0.006]	–0.123 [0.078]
Duration	0.004 [0.007]	0.005 [0.004]	0.003*** [0.001]
Trading volume	–	0.005 [0.004]	0.008 [0.006]
Number of trades	–	–0.016*** [0.005]	–0.001 [0.004]
Fixed effects			
Firm-quarter	Yes	Yes	Yes
Observations	2,397	2,283	2,435
Clusters (issuers)	50	50	46
R ²	0.36	0.37	0.39

(a) Table VI: Secondary market trading

Table VII
Reaching for Yield: Time-Series Variation

This table reports results from panel regressions of secondary market activity of insurance companies in investment-grade corporate bonds. The dependent variable is the log quarterly change in the value of all insurance companies' holdings of a bond (maturing bonds are excluded). The sample is constrained to investment-grade issues (NAIC categories 1 and 2). The coefficient on yield is allowed to vary by quarter. Standard errors are clustered by issuer and are reported in a separate column. Superscript "r" indicates NBER recession quarters. *** and ** indicate statistical significance at the 1% and 5% level, respectively.

Dependent variable	Log Change in the Value of Insurance Companies' Total Holdings Since the Previous Quarter	
	Coeff.	SE
Interaction terms (yield spread × quarter)		
2004:Q2	0.014	0.012
2004:Q3	0.005	0.011
2004:Q4	0.015	0.015
2005:Q1	0.025**	0.010
2005:Q2	–0.002	0.005
2005:Q3	0.035***	0.010
2005:Q4	0.032***	0.008
2006:Q1	0.001	0.006
2006:Q2	0.050***	0.008
2006:Q3	0.019**	0.009
2006:Q4	0.034**	0.007
2007:Q1	0.060***	0.006
2007:Q2	0.018	0.006
2007:Q3 ^r	–0.031***	0.006
2007:Q4 ^r	0.005	0.007
2008:Q1 ^r	0.008	0.007
2008:Q2 ^r	0.002	0.005
2008:Q3 ^r	–0.004	0.008
2008:Q4 ^r	0.009**	0.003
2009:Q1 ^r	0.000	0.002
2009:Q2 ^r	0.003	0.006
2009:Q3	0.045***	0.014
2009:Q4	0.012	0.012
2010:Q1	0.002	0.011
2010:Q2	0.020***	0.007
2010:Q3	0.012	0.009
Firm-quarter fixed effects:	Yes	
Observations	6,486	
R ²	0.27	

(b) Table VII: Time-series variation

6.7 Table VIII: Bond performance return analysis and Table IX: Bond performance downgrades

Read:

- Table VIII compares returns for bonds with high versus low insurance holdings at issuance.
- Bonds with high insurance holdings have negative and significant alphas in several factor models.
- Table IX shows that higher insurance holdings predict higher future downgrade probability and worse future rating changes during the benchmark period.

Interpretation: These tests reject the simplest skill story. If insurers were merely selecting underpriced bonds, high-insurance bonds should outperform or at least not be downgraded more. Instead, high-insurance bonds appear riskier and do not deliver superior returns.

Economic message: The extra yield is compensation for hidden risk, not free alpha. Reaching exposes insurers to downside risk that ratings do not fully capture.

Table VIII
Bond Performance: Return Analysis

This table examines bonds' secondary market performance for 2004 to 2007. The test focuses on newly issued investment-grade bonds. The dependent variable is monthly excess return (returns in excess of the risk-free rate) measured in bps. Returns are constructed using prices from noncanceled transactions reported in TRACE. We exclude any bond trades below \$1 million to make sure that we capture institutional rather than retail transactions. Monthly returns are computed using the median trade price the last day of the month and factor in bond coupons. For a given month the sample includes bonds acquired within a one-year window. We report equal-weighted excess returns for the portfolio with the highest (above-median) and lowest (below-median) insurance holdings measured at issuance. The risk-free rate and market return are from Ken French's website. *Default premium* is the difference in returns between investment-grade and high-yield corporate bonds (the Lehman/Barclays index). *Term spread* is the difference in returns between the five-year Treasury bond and the three-month Treasury bill. *Liquidity factor* is the return factor of Pastor and Stambaugh (2003). Robust standard errors are reported in brackets below coefficients ***, **, and * indicate statistical significance at 1%, 5%, and 10% level, respectively.

Bond Portfolio	High Insurance Holdings			Low Insurance Holdings		
	(1)	(2)	(3)	(4)	(5)	(6)
	Fama and French (1989)			Fama and French (1989)		
Intercept ("α")	-36.0* [18.4]	-49.7*** [18.4]	-67.9** [20.3]	-8.8 [8.8]	-16.5 [10.1]	-22.4* [11.8]
Stock market excess return	—	0.211** [0.086]	—	—	0.120** [0.047]	—
Default premium	0.302* [0.176]	0.273 [0.171]	0.224 [0.159]	0.083 [0.082]	0.066 [0.074]	0.051 [0.073]
Term spread	0.364*** [0.074]	0.414*** [0.060]	0.403*** [0.064]	0.297*** [0.045]	0.324*** [0.040]	0.321*** [0.043]
Liquidity factor	—	—	0.184*** [0.061]	—	—	0.059* [0.031]
Observations	48	48	48	48	48	48
R ²	0.27	0.39	0.49	0.47	0.58	0.61
Mean excess return	-21.1			2.2		
SD	150.8 (Diff.: 62.6***)			88.2		
Mean return – IG index	-28.9			-7.6		

(a) Table VIII: Return analysis

Table IX
Bond Performance: Downgrades

This table examines ratings changes for newly issued corporate bonds. The sample is constrained to investment-grade issues (NAIC categories 1 and 2). The aggregate credit rating of a bond is based on ratings by S&P, Moody's, and Fitch. If a bond has two ratings, the lower is used, and if it has three ratings, the median is used. *Downgrade indicator* is equal to one if the (aggregate) credit rating fell by at least one notch over the specified time interval, and zero otherwise. *Rating change* is the future (aggregate) rating minus the (aggregate) rating at issue and ranges from -1 to +3. Robust standard errors, clustered by bond issue, are reported in brackets below coefficients ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

Dependent Variable	Benchmark (2004:Q3 –2007:Q2)			Crisis (2007:Q3 –2010:Q4)
	Downgrade Indicator One Year Out (1)	Downgrade Indicator Years Out (2)	Ratings Change Two Years Out (3)	Downgrade Indicator One Year Out (3)
Insurance companies' holdings at issue	0.108** [0.055]	0.231*** [0.073]	-0.833** (0.417)	0.208 [0.180]
NAIC category 2 (BBB+, BBB, BBB-)	0.070	0.063	-0.393	-0.175* [0.089]
Constant	0.096 [0.043]	0.115 [0.061]	(0.487) (0.327)	0.095 [0.089]
Observations	901	901	901	474
Clusters (issuers)	155	155	155	0.185
R ²	0.014	0.022	0.025	0.065

(b) Table IX: Downgrades

6.8 Table X: Governance and reaching for yield and Table XI: Reaching for yield and insurers' performance

Read:

- Table X shows that lower regulatory capital surplus is associated with higher average yield spreads in insurers' investment-grade purchases.
- Stronger blockholder ownership is associated with lower portfolio yield spreads.
- Table XI shows that insurers with higher pre-crisis average yield spreads have worse stock returns during the financial crisis.
- The relation remains after controls for TARP receipt, stock beta, and stock volatility.

Interpretation: Table X connects reaching to incentives and governance; Table XI connects reaching to realized downside risk. Together they show that reaching is not just a portfolio-style preference; it is linked to institutional constraints and later losses.

Economic message: Weak incentives and binding capital rules can push firms toward hidden risk. The risks are realized when market conditions deteriorate.

Table X
Governance and Reaching for Yield

This table reports results from cross-sectional regressions of purchases of insurance companies NAIC categories 1 and 2 (i.e., investment-grade) corporate bonds. We look at purchases of bonds issued during the period 2003Q4 to 2007Q2. Each observation is one insurance company and year. The dependent variable, *Yield spread*, is the average yield spread (the difference between bonds acquired by a given insurance company and duration-matched Treasury), averaged for a firm-year. When a firm acquires stakes in less than three bonds in a given year, it is dropped. *Regulatory capital surplus* is capital surplus normalized by an insurer's assets. *Blockholder ownership* measures the fraction of outstanding shares held by blockholders. *Public firm* is a dummy indicator whether the firm is publicly traded. *Leverage* is the ratio of liabilities to equity, and is capped at 20. *Size* is the log of assets (book value). Firms with less than \$10 million in asset value are dropped. Additionally, firms with less than \$10 million in market capitalization are dropped. Columns (2), (3), (5), and (6). Robust standard errors, clustered by firm, are reported in brackets. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

Dependent Variable	Average Yield Spread, Corporate Bonds in NAIC 1-2			Average Yield Spread, Corporate Bonds in NAIC 1		
	(1)	(2)	(3)	(4)	(5)	(6)
Insurance firms characteristics						
Regulatory capital surplus	-109.3*** [29.6]	-	-116.2* [70.4]	-93.1*** [26.6]	-	-114.2* [42.0]
Blockholder ownership	-	-84.9*** [18.2]	-54.4** [20.0]	-	-73.4*** [16.1]	-49.6* [19.8]
Public firm	1.58 [5.37]	-	-	2.63 [4.88]	-	-
Leverage	-	-	12.2 [30.8]	-	-	-1.1 [25.4]
Size	-	-	2.44 [2.86]	-	-	-0.2 [2.3]
Observations	379	177	158	373	175	155
R ²	0.09	0.17	0.29	0.09	0.14	0.11

(a) Table X: Governance and reaching

Table XI
Reaching for Yield and Insurance Companies' Performance

This table reports results from regressing insurance companies' stock returns during the financial crisis (June 2008 to February 2009) on their precrisis portfolio choices. Each observation in the analysis corresponds to a different insurance firm (this is a pure cross section). In columns 1 and 3, the dependent variable is the raw return; in column (2), the dependent variable is the excess return calculated using a market model with β s estimated based on 2002 to 2010 data. *Average yield spread* is the value-weighted average spread for bonds acquired by a given insurance company at issuance. The bond sample is constrained to NAIC category 1 corporate bonds issued between 2004:Q3 and 2007:Q2. *TARP recipient* indicates three insurance companies that received support under the Federal Reserve's Troubled Asset Relief Program in the fall of 2008. *Stock β* and *Stock volatility* are estimated using daily data for the 2001 to 2011 period. Robust standard errors are reported in brackets. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

Dependent Variable	Return (1)	Excess Return (2)	Return (3)
Insurance firms characteristics			
Average yield spread	-0.824*** [0.191]	-0.359* [0.188]	-0.399* [0.184]
TARP recipient	-	-	-0.058 [0.068]
Stock β	-	-	-0.114 [0.070]
Stock volatility	-	-	-4.76** [1.435]
Observations	67	67	67
R ²	0.25	0.06	0.52

(b) Table XI: Crisis performance

6.9 Figure 6: Insurance companies' stock price performance during the financial crisis and Figure 7: Reaching for yield and bond issuance

Read:

- Figure 6 plots insurer crisis stock returns against their pre-crisis average yield spreads in NAIC 1 bonds.
- The negative relation shows insurers that reached more suffered larger equity losses in the crisis.
- Figure 7 overlays insurer reaching with the share of NAIC 2 issuance by high-CDS firms.
- High-CDS BBB issuers account for more issuance when insurers reach more for yield.

Interpretation: Figure 6 turns portfolio choice into firm-level risk exposure. Figure 7 suggests a credit-supply channel: when insurers demand high-yield bonds within the same rating bucket, risky issuers respond by issuing more.

Economic message: Reaching can affect both investor balance sheets and issuer financing decisions, linking micro portfolio incentives to aggregate credit cycles.

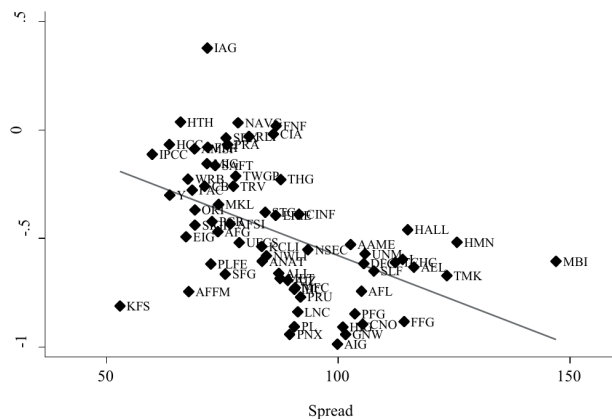


Figure 6. Insurance companies' stock price performance during financial crisis. The figure plots stock returns for the period June 2008 to February 2009 relative to their bond invest

(a) Figure 6: Crisis stock performance

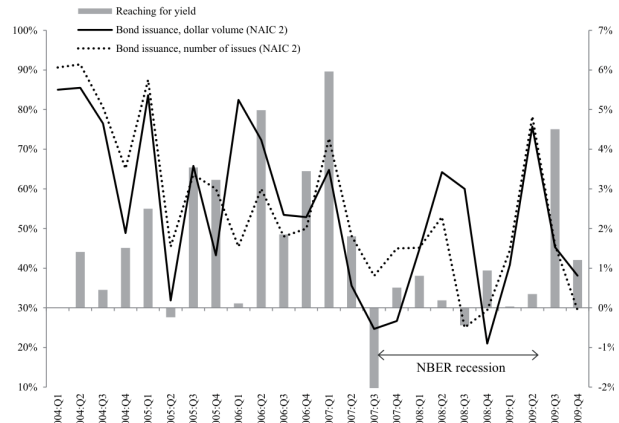


Figure 7. Reaching for yield and bond issuance. The figure presents insurance companies' preference for high-yielding bonds (bars, right axis) and the relative bond issuance of high CDS corporates (lines, left axis) within NAIC category 2 (ratings between BBB- and BBB+). The

(b) Figure 7: Reaching and issuance

7 Interpretive synthesis

The paper's logic is a three-step chain. First, ratings-based regulation creates discrete risk categories. Second, within those categories, market prices reveal additional risk through yield and CDS spreads. Third, insurers have incentives to buy the higher-yield bonds that satisfy the same rating constraint. This is reaching for yield.

The evidence supports this chain at multiple levels. Security-level holdings show insurers tilt toward high-yield and high-CDS bonds. Time-series tests show the tilt is strongest in credit booms. Return and downgrade tests show the tilt does not produce alpha. Insurer-level tests show the tilt is stronger where capital and governance incentives are weaker. Finally, issuance tests show that risky firms appear to issue more when reaching is stronger.

The broader lesson is that risk regulation is only as good as the risk metric it uses. If regulation treats two bonds as equivalent while the market prices them differently, regulated institutions may optimize against the regulatory measure rather than against true risk.

8 Short summary

- The paper studies reaching for yield in the corporate bond market.
- Insurance companies are the focus because they are large bond investors and face rating-based capital requirements.
- Insurers prefer investment-grade bonds because capital charges are lower for safer NAIC categories.
- Conditional on being in the same rating category, insurers overweight higher-yield and higher-CDS bonds.
- This reaching pattern is strongest in 2004–2007, disappears during the crisis, and reappears after the crisis.
- Secondary-market trading confirms that insurers actively add more high-yield investment-grade bonds in good times.
- High-insurance bonds do not earn positive abnormal returns; many have negative alphas.
- High-insurance holdings predict future downgrades.

- Insurers with low regulatory capital surplus and weak blockholder governance reach more.
- Insurers that reached more before the crisis performed worse during the crisis.
- High-CDS BBB issuers appear to issue more bonds when reaching is stronger.
- The paper’s main conclusion is that imperfect risk benchmarks can create hidden risk-taking incentives.

9 Conclusion

9.1 Core conclusion

Becker and Ivashina provide security-level evidence that insurance companies reach for yield in corporate bond markets. They prefer safer rating buckets overall, but within a rating bucket they tilt toward bonds with higher yields and higher CDS spreads. This is exactly the behavior predicted when regulation and delegated management use ratings as incomplete risk measures.

9.2 Economic intuition

Ratings summarize some default risk but not all priced risk. A high-yield BBB bond and a low-yield BBB bond may impose similar capital charges even though markets view the high-yield bond as riskier. An insurer seeking higher portfolio yield can exploit this by buying the riskier bond without moving to a lower NAIC category. The behavior raises apparent income in good times but creates hidden exposure to downturns.

9.3 What the paper establishes

The paper establishes that reaching for yield is visible in actual institutional holdings, is stronger in expansions, is related to capital and governance incentives, and is not explained by superior bond selection. It also shows that reaching has consequences for insurers and possibly for corporate issuance.

9.4 What the paper does not fully establish

The analysis is not a randomized experiment. It cannot prove that all high-yield holdings are caused by regulation, nor can it fully separate every unobserved preference or liability-driven motive. The issuance evidence is suggestive rather than definitive because it is based on a short and turbulent time series. Still, the combined evidence strongly supports the view that imperfect risk metrics distort investment incentives.

9.5 Final takeaway

The paper’s enduring contribution is to show how financial institutions can take risk inside the rules. Regulation that relies on coarse ratings can push investors toward hidden risk rather than away from risk. This mechanism helps explain why credit booms can involve apparently safe assets that later generate large losses.